The cost of parametric Gröbner basis computation over a rational function field

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Abstract

Inverse kinematics can be posed so that the target pose is adjoined to the coefficient field rather than substituted into the polynomial ring. A single Gröbner basis over $\mathbb{Q}(\boldsymbol{p})$ then answers every pose, which is what permits an offline solve and generated runtime code. In the implementation studied here that computation completes in milliseconds with two adjoined parameters and has not been observed to complete with three. Profiling attributes approximately 86 % of the running time to the polynomial greatest common divisor invoked when coefficients are reduced to lowest terms, which suggests that a modular gcd would remove the limitation. This report tests that suggestion by repeating the computation over $\mathbb{F}_p(\boldsymbol{p})$, where coefficient arithmetic is a single machine-word operation and coefficients do not grow. The three-parameter problem fails to terminate over $\mathbb{F}_p$ as well. An independent measurement of the subresultant gcd over both fields, on dense trivariate operands ranging from 35 to 969 terms, gives fitted cost exponents of 3.79 over $\mathbb{F}_p$ and 3.81 over $\mathbb{Q}$, with a ratio between the fields that rises to and settles near 13. The choice of coefficient field therefore governs a bounded multiplicative constant while the growth in operand size governs the exponent. Since the operands reached during a single reduction of the three-parameter run number 498 terms, a value interior to the measured range, a modular gcd formed by computing the existing remainder sequence over several primes cannot make the problem tractable, and only a gcd based on evaluation and interpolation addresses the measured cost.

1 Introduction

The inverse kinematics of a serial manipulator with revolute joints becomes a polynomial system under the tangent half-angle substitution $t_i = \tan(q_i/2)$, and the resulting zero-dimensional ideal can be solved by the eigenvalue method of Möller and Stetter [10]. Classical treatments of the general $6R$ arm reduce the problem to a univariate resultant of degree sixteen [8, 13, 9]. The system studied here takes a different route. The target pose is adjoined to the coefficient field, so that the Gröbner basis is computed once for the manipulator and evaluated at each requested pose, and the action matrices of the quotient algebra are emitted as source code.

The cost of that formulation is the subject of this report. Writing N for the number of actuated joints and P for the number of adjoined pose coordinates, the implementation completes the case $P = N = 2$ in between 6 ms and 33 ms, and has produced no result for $P = N = 3$ after 900 s. An instrumented run of the three-parameter problem attributes approximately 86 % of elapsed time to the polynomial gcd invoked during normalisation of the coefficients.

Two explanations are consistent with that attribution, and they recommend different work. The first locates the cost in arithmetic on rational coefficients, whose numerators and denominators grow through the subresultant remainder sequence; on this account a modular gcd, computing images over several primes and reconstructing by the Chinese remainder theorem, would recover most of the running time. The second locates it in the number of terms of the parameter polynomials, a quantity fixed by the problem and the monomial order and independent of how coefficients are represented. Deciding between the two accounts before committing to an implementation is the purpose of the experiment described below.

The method is a controlled substitution of the coefficient field, replacing $\mathbb{Q}$ by a prime field in which coefficients cannot grow, together with a control that detects an invalid substitution; Section 3 sets this out. Section 4 reports that the three-parameter problem fails to terminate over the prime field, and localises the failure to a single reduction rather than to an accumulating basis. Section 4.3 measures the subresultant gcd in isolation over both fields, across a range of operand sizes that covers the sizes reached by the failing run. Section 5 separates the multiplicative effect of the coefficient field from the exponent in operand size and states what each class of modular gcd can address.

2 Preliminaries

2.1 Notation

Let k be a field and $k[\boldsymbol{t}] = k[t_1, \dots, t_N]$ the polynomial ring in the joint variables. For a monomial order $\prec$ and nonzero $f \in k[\boldsymbol{t}]$, $\mathrm{LM}(f)$ denotes the leading monomial of f, and for an ideal I the initial ideal is $\mathrm{in}(I) = \langle \mathrm{LM}(f) : f \in I \setminus \{0\} \rangle$. A finite $G \subset I$ is a Gröbner basis when $\mathrm{in}(I) = \langle \mathrm{LM}(g) : g \in G \rangle$. The quotient $A = k[\boldsymbol{t}]/I$ is a k-vector space with basis the monomials outside $\mathrm{in}(I)$, called the standard monomials, and I is zero-dimensional exactly when that basis is finite, in which case $\dim_k A$ bounds the number of points of $V(I)$ over $\bar{k}$ and equals it when I is radical. We refer to [5] throughout for these facts and to [3] for the completion algorithm.

Saturation is written $I : h^\infty$. The half-angle substitution introduces denominators $1 + t_i^2$, and clearing them attaches to $V(I)$ the loci $t_i = \pm\sqrt{-1}$, which correspond to no configuration of the manipulator. Saturating by $h = \prod_i (1 + t_i^2)$ removes them, so that $\dim_k A$ counts configurations alone.

2.2 The parametric formulation

Definition 1 (Parametric position problem)**.** Let $\Phi : \mathbb{A}^N \to \mathbb{A}^3$ be the rational forward map of a manipulator with N actuated joints, written over the coefficient field k, with numerator components Φ_i and common denominator D. Let $C \subseteq \{1, 2, 3\}$ with $|C| = P$ index the constrained tool coordinates, and adjoin indeterminates $\boldsymbol{p} = (p_1, \dots, p_P)$ to k. The parametric position ideal is

$$I \;=\; \Big\langle\, \Phi_{c_j}(\boldsymbol{t}) - D(\boldsymbol{t})\, p_j \;:\; j = 1, \dots, P \,\Big\rangle : h^\infty \;\subseteq\; k(\boldsymbol{p})[\boldsymbol{t}].$$

Coefficients are elements of $k(\boldsymbol{p})$, represented as a quotient of two polynomials in $\boldsymbol{p}$ over k and reduced to lowest terms after every arithmetic operation. Reduction requires a polynomial gcd in $k[\boldsymbol{p}]$, computed by a subresultant remainder sequence in the sense of Collins [4]. Without it the representations grow without bound under the repeated additions and multiplications of the completion algorithm, so the reduction is a requirement of the formulation rather than an economy.

2.3 The admissible number of parameters

The two results below restrict attention to $P = N$ and identify the three-parameter case as the smallest spatial instance.

Proposition 2. *Let I be as in Definition 1 with $P < N$. Then every irreducible component of $V(I)$ has dimension at least $N - P$, and I fails to be zero-dimensional unless $V(I) = \emptyset$.*

Proof. The ideal before saturation is generated by P elements, so by Krull's height theorem [6, Ch. 10] every minimal prime over it has height at most P, whence every component of its zero set has dimension at least $N - P$. Saturation deletes components and leaves the dimension of those that survive unchanged, so the bound persists. The degenerate case $V(I) = \emptyset$ arises when saturation deletes every component. □

Proposition 3. *Let I be as in Definition 1 with $\boldsymbol{p}$ algebraically independent over k, and suppose $P > N$. If the Zariski closure of the image of Φ has dimension at most N, then $I = \langle 1 \rangle$.*

Proof. Since $p_1, \ldots, p_P$ satisfy no nontrivial polynomial relation over k, the point $\boldsymbol{p}$ is a generic point of $\mathbb{A}^P$. The attainable tool positions lie in a constructible set whose closure $\overline{\operatorname{im}\Phi}$ has dimension at most $N < P$, hence is a proper closed subset of $\mathbb{A}^P$ and is contained in the zero set of some nonzero $F \in k[\boldsymbol{p}]$. Independence gives $F(\boldsymbol{p}) \neq 0$, so no $\boldsymbol{t}$ satisfies the generators and the ideal is the unit ideal. □

Corollary 4. *A nonempty finite solution set requires $P = N$.*

Remark 5. The implementation uses Corollary 4 as a precondition test, rejecting $P \neq N$ by counting before any completion is attempted. Under this restriction the three-parameter case is the smallest one in which a manipulator positions a tool in space.

3 Experimental design

3.1 Substitution of the coefficient field

The discriminating measurement replaces $\mathbb{Q}$ by the prime field $\mathbb{F}_p$ with $p = 2^{31} - 1$, holding fixed the algorithm, the monomial order, the generators, the saturation and the number of adjoined parameters. In $\mathbb{F}_p$ an addition or multiplication is one machine-word operation followed by a reduction, and coefficients do not grow, so any cost surviving the substitution is independent of coefficient size. The substitution therefore separates the two accounts of Section 1 directly.

The field $\mathbb{F}_p(\boldsymbol{p})$ is implemented as a quotient of two polynomials over $\mathbb{F}_p$, normalised in the same three steps used for $\mathbb{Q}(\boldsymbol{p})$: division by the monomial content, division by the polynomial gcd, and scaling to a monic denominator. The remainder of the library is templated on the coefficient type and is used without modification, so the two runs execute the same instruction sequence up to the arithmetic itself.

3.2 Validity control

Reduction modulo a prime preserves the structure of a Gröbner basis for all but finitely many primes. The exceptional primes are those for which the reduced basis acquires a different initial ideal, and detecting them is the standard obstacle in modular Gröbner methods [1]. We therefore record, for every completed run, the multiset of leading monomials of the reduced basis.

Remark 6. Agreement of the initial ideals between the two fields is a necessary condition for the substitution to be valid on a given input. Disagreement would indicate an unlucky prime and would invalidate a comparison of running times on that input, the two runs having completed different ideals.

3.3 Systems

Three systems are measured, each posed as in Definition 1 with one adjoined coordinate per joint, so that $P = N$ throughout in accordance with Corollary 4. The first is a planar two-link arm with two revolute joints about z and unit links, constrained in x and y, which has two solutions. The second is the shoulder and elbow of an anthropomorphic arm, obtained by removing its base joint in the manner of Pieper's decoupling [12] and constrained by a radius and a height; it also has two solutions. The third is the anthropomorphic arm itself, a base yawing about z followed by two joints pitching about y, with unit links and all three coordinates constrained. Completing that system over $\mathbb{Q}$ at a fixed rational pose establishes independently that it has four solutions. The first two systems act as controls in the sense of Remark 6.

3.4 Instrumentation

Three quantities are recorded beyond wall-clock time. The completion is timed separately from the construction of the forward map and the residuals, since the latter also performs coefficient arithmetic and would otherwise be conflated with the former. The main loop of the completion algorithm reports the number of pairs processed, the current basis size, the pending pair count and the largest basis polynomial. The normalisation of $k(\boldsymbol{p})$ reports the largest numerator or denominator it has encountered, measured in terms.

3.5 Environment

Measurements were taken on an Intel Core i7-10870H at 2.20 GHz running Linux 6.8, compiled with g++ 11.4.0 at `-O2`, with GMP providing arbitrary-precision rational arithmetic. In the gcd measurements of Section 4.3, points whose operands have total degree at most eight are the median of five runs and larger points are single runs; the quantity of interest spans five orders of magnitude and is insensitive to run-to-run variation at that scale.

4 Results

4.1 Controls

Table 1 reports the two-parameter systems. Basis size, quotient dimension, largest basis polynomial and initial ideal agree exactly between the fields, so by Remark 6 the prime is not unlucky for these inputs. The substitution reduces the completion time by a factor of approximately four.

Table 1: Two-parameter systems, both fields. Times in seconds. *Setup* covers construction of the rational forward map and the residuals; *completion* covers saturation and the Gröbner basis. The initial ideal is given by the leading monomials of the reduced basis.

System	Field	Setup	Completion	Basis	$\dim_k A$	Max terms	Initial ideal
Planar 2R	$\mathbb{F}_p(x, y)$	0.000	0.009	2	2	3	$\langle t_2^2,\, t_1\rangle$
Planar 2R	$\mathbb{Q}(x, y)$	0.001	0.031	2	2	3	$\langle t_2^2,\, t_1\rangle$
Reduced 2R	$\mathbb{F}_p(r, z)$	0.000	0.008	2	2	3	$\langle t_2^2,\, t_1\rangle$
Reduced 2R	$\mathbb{Q}(r, z)$	0.001	0.033	2	2	3	$\langle t_2^2,\, t_1\rangle$

4.2 The three-parameter system

The anthropomorphic 3R system with all three coordinates constrained produced no result over $\mathbb{F}_p(x, y, z)$ within 400 s, nor over $\mathbb{Q}(x, y, z)$ within 900 s. Construction of the forward map and the residuals completed in under 0.05 s in both fields, so the whole of the elapsed time is attributable to the completion.

Instrumenting the main loop localises the failure. Over $\mathbb{F}_p(x, y, z)$ the run enters the loop with four basis elements, six pending pairs and a largest basis polynomial of fourteen terms, and processes one pair. No second pair was processed in the subsequent 300 s. The run is therefore inside a single reduction, and the failure is not an accumulation of many basis elements.

Instrumenting the normalisation identifies what grows during that reduction. The largest parameter polynomial appearing as a numerator or denominator reaches 460 terms after 93 s and 498 terms after 129 s, with intervals of tens of seconds between consecutive normalisations. Coefficients in $\mathbb{F}_p$ do not grow, so these intervals measure the cost of a gcd in $\mathbb{F}_p[x, y, z]$ on operands of that size. This observation motivates the measurement of the next section and fixes $m \approx 500$ as the operating point of interest.

4.3 The subresultant gcd in isolation

Table 2 and Figure 1 report the cost of a single gcd outside the pipeline. Two dense trivariate polynomials of equal total degree are formed with a planted common factor, so that the remainder sequence runs to full length rather than terminating early on coprime inputs. The same integer coefficients are used over both fields.

Table 2: One subresultant gcd on dense trivariate operands with a planted common factor. Here m is the number of terms per operand and t the running time in seconds.

Degree	m	gcd terms	t over $\mathbb{F}_p$	t over $\mathbb{Q}$	Ratio
4	35	10	0.000505	0.005692	11.3
6	84	20	0.008278	0.099993	12.1
8	165	35	0.114564	1.519231	13.3
10	286	56	1.073065	14.463953	13.5
12	455	84	6.554416	91.276337	13.9
14	680	120	31.064855	–	–
16	969	165	124.013371	–	–

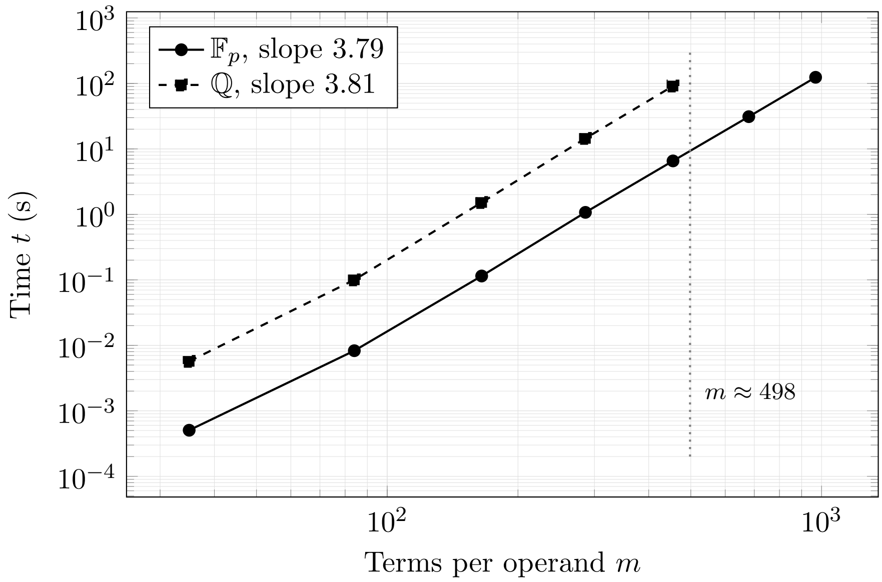

Figure 1: Cost of one subresultant gcd against operand size, both axes logarithmic. The two series are parallel, so the coefficient field contributes a multiplicative constant and leaves the exponent unchanged. The vertical line marks the operand size observed during the failing three-parameter run, which lies inside the measured range.

Fitting $\log t = \alpha \log m + \beta$ by least squares over the points of Table 2 gives $\alpha_{\mathbb{F}_p} = 3.79$ and $\alpha_{\mathbb{Q}} = 3.81$. The ratio $t_{\mathbb{Q}}/t_{\mathbb{F}_p}$ rises from 11.3 at $m = 35$ and settles near 13.9 at $m = 455$.

5 Analysis

5.1 Separation of the two effects

The controls of Table 1 establish that the substitution of $\mathbb{F}_p$ for $\mathbb{Q}$ preserves the computation on the two-parameter inputs, by the criterion of Remark 6. The three-parameter system then fails to terminate in a setting where coefficient arithmetic is a machine-word operation and coefficients do not grow, which is inconsistent with the first of the two accounts as an explanation of the failure.

Table 2 quantifies what remains. The two fitted exponents agree to within 0.02, so the coefficient field affects the constant β and leaves α unchanged, which appears in Figure 1 as two parallel lines. The exponent $\alpha \approx 3.8$ is a property of the subresultant remainder sequence applied to multivariate operands: the sequence performs a number of pseudo-divisions growing with the degree in the main variable, each pseudo-division is a polynomial multiplication and exact division over the remaining variables, and the content computations recurse into those variables in turn [4, 2]. The measured α is consistent with that structure and is present over both fields.

5.2 Consequences for modular gcd algorithms

The term *modular gcd* covers two constructions with different consequences here. The first computes the same subresultant remainder sequence over several primes and reconstructs the result by the Chinese remainder theorem, with a trial division to certify the candidate [2]. It addresses coefficient growth and leaves the sequence itself intact, so its benefit is bounded by the ratio measured in Table 2, approximately a factor of 13. Applied to a computation that has not completed in 900 s, a factor of 13 does not produce a usable result.

The second reduces the multivariate problem to univariate gcds at sampled points and recovers the answer by interpolation, in the dense form due to Brown [2] or the sparse form due to Zippel [14], with the Hensel-based variant of Moses and Yun [11] a further alternative. Such a construction replaces the remainder sequence and therefore acts on α rather than on β, which is what the measurement identifies as the governing quantity.

5.3 The operating point

Interpolating Table 2 between $m = 455$ and $m = 680$ at the exponent $\alpha = 3.79$ places a single gcd at $m = 498$ at approximately 9 s over $\mathbb{F}_p$ and, applying the measured ratio, approximately 125 s over $\mathbb{Q}$. Both figures agree with the intervals between consecutive normalisations reported in Section 4.2. Since $m = 498$ lies inside the range covered by Table 2, the estimate requires no extrapolation.

6 Threats to validity

The experiment establishes that a faster gcd is necessary for the three-parameter problem without establishing that it is sufficient. The number of reductions the completion requires is unknown, the run having never terminated, and a gcd with a lower exponent would remove the present obstruction without guaranteeing termination in useful time. Settling the question would require either a completed run or an independent bound on the size of the reduced basis over $\mathbb{Q}(\boldsymbol{p})$.

The operands of Table 2 are dense of their total degree, whereas the parameter polynomials arising in the pipeline need not be. Density fixes the relation between degree and term count and makes m an unambiguous size measure, at the cost of favouring dense interpolation over the sparse method of [14] in any comparison of the two. The separation of α from β, which is the conclusion drawn, is unaffected.

One prime was used for the substitution. The control of Remark 6 verifies that it is not unlucky on the two-parameter inputs, and no such verification is available on the three-parameter input, which does not complete. An unlucky prime would be expected to alter the initial ideal and hence the course of the computation, and would not be expected to convert a terminating computation into a non-terminating one, the failure being localised to a single reduction whose operands grow for reasons independent of the prime.

The statements that the three-parameter runs do not complete are bounded observations at 400 s and 900 s rather than proofs of non-termination. The instrumentation of Section 4.2, showing one pair processed and none thereafter, is the stronger evidence and is what the analysis relies upon.

Finally, all measurements concern one implementation of Buchberger's algorithm with the normal selection strategy and the graded reverse lexicographic order. A completion algorithm of the F4 family [7], which replaces sequential reduction by row reduction of a large matrix, would change the number and the shape of the coefficient operations, and has not been measured.

7 Conclusions

The three-parameter parametric position problem does not complete over $\mathbb{F}_p(x, y, z)$ within 400 s, so its cost is not attributable to arithmetic on rational coefficients, and the first of the two accounts of Section 1 is rejected as an explanation of the limitation. The subresultant gcd on dense trivariate operands instead costs $\Theta(m^\alpha)$ with $\alpha \approx 3.8$ measured over both fields, the coefficient field contributing a bounded multiplicative factor of approximately 13. The operands reached within one reduction of the three-parameter run number 498 terms, a value interior to the measured range, which places a single gcd at approximately 9 s over $\mathbb{F}_p$.

It follows that a modular gcd formed by computing the existing remainder sequence over several primes is bounded by that factor of 13 and would not render the three-parameter problem tractable, whereas a gcd based on evaluation and interpolation acts on the exponent and is the construction the measurement points to. Whether it also suffices is left open, for the reason given in Section 6. For the implementation as it stands, $P = 2$ is the working limit, and three-joint arms are better obtained by the decoupling of [12], which reduces such an arm to a two-parameter problem completed in 25 ms.

A Reproduction

From the repository root:

```
g++ -std=c++17 -O2 doc/experiments/field_cost.cpp -o field_cost \
  -Ivarietas_core/include -Ivarietas_codegen/include \
  -Ivarietas_kinematics/include -Idoc/experiments \
  -I/usr/include/eigen3 -lgmpxx -lgmp

./field_cost planar      # two-parameter control, both fields
./field_cost reduced     # two-parameter control, both fields
./field_cost spatial-fp  # three parameters over F_p; does not terminate
./field_cost spatial-q   # three parameters over Q;   does not terminate

g++ -std=c++17 -O2 doc/experiments/gcd_cost.cpp -o gcd_cost \
  -Ivarietas_core/include -Ivarietas_codegen/include \
  -Idoc/experiments -I/usr/include/eigen3 -lgmpxx -lgmp

./gcd_cost               # Table 2 and the fitted exponents
```

Adding `-DVARIETAS_EXPERIMENT_TRACE` to the first build reports the largest parameter polynomial encountered during normalisation, which is the source of the figure $m = 498$. The pair-queue statistics of Section 4.2 were obtained with a temporary instrumented copy of `varietas_core/include/varietas/core/ideal/buchberger.hpp` printing the basis size, the pending pair count and the largest basis polynomial on each pass through the main loop.

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